2 **(a)** State *Newton's third law of motion*.

....

....

.... [1]

(b) In a track and field event, a high jumper jumps over a horizontal bar and lands on a thick mattress. Fig. 2.1 shows the instant where she is just clearing the bar at her maximum height. The high jumper is considered as a point mass labelled P.

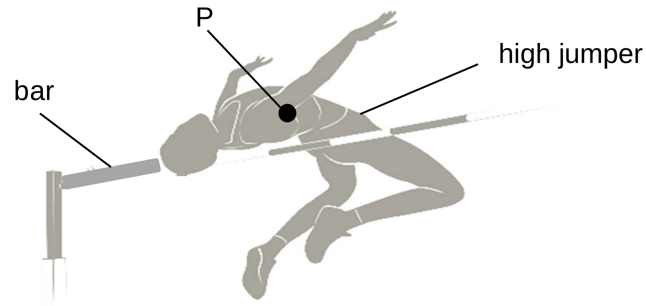

Fig. 2.1

(i) On Fig. 2.1, draw a labelled force diagram to show the force(s) acting on her at P. Make it clear the difference in magnitude of the force(s) if any. [1]

(ii) Fig. 2.2 shows the instant she lands on the mattress.

Fig. 2.2

On Fig. 2.2, draw labelled force diagrams to show the forces on her and on the mattress for the instant she landed. Make it clear which forces are action-reaction pairs, and which forces are different in magnitude. [3]

(c) Fig. 2.3 shows the variation with time t of the impact force F on her upon hitting the mattress. She has a mass of 50 kg and she comes to rest without rebounding at t = 1.2 s.

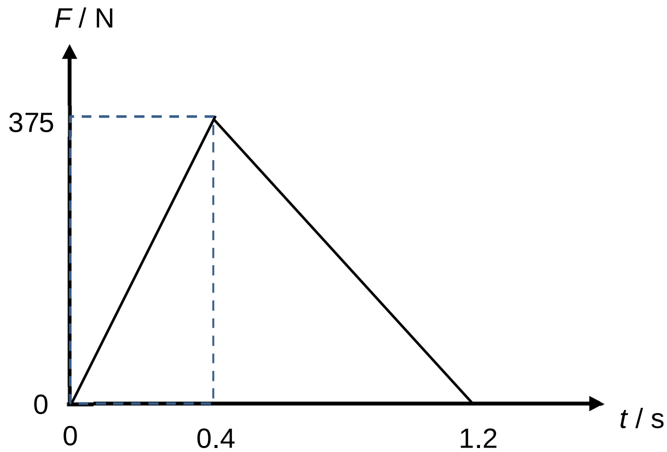

Fig. 2.3

(i) Determine the speed at which she hits the mattress.

speed = ... m s^{-1} [2]

(ii) In another attempt, she hits the high jump bar horizontally with a speed of 1.5 m s^{-1}. The bar has a mass of 3.0 kg and the collision is elastic.

Determine the speed of the bar immediately after collision.

speed = ... m s^{-1} [3]

3 **(a)** Define *gravitational potential* at a point.